Project Deadline - 2

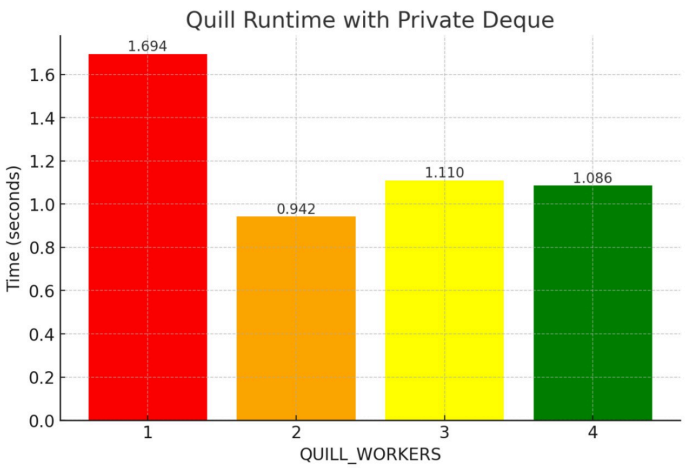
Quill Runtime with Private Deque -

Every worker has a private deque of its own which does not require locks.

Execution Time -

```
~/Desktop/SEM6/PRMP/Project/A2 (1.772s)
QUILL_WORKERS=1 ./nqueens 12
OK
NQueens(12) Time = 1.693965sec

~/Desktop/SEM6/PRMP/Project/A2 (1.021s)
QUILL_WORKERS=2 ./nqueens 12
OK
NQueens(12) Time = 0.942360sec

~/Desktop/SEM6/PRMP/Project/A2 (1.192s)
QUILL_WORKERS=3 ./nqueens 12
OK
NQueens(12) Time = 1.110115sec

~/Desktop/SEM6/PRMP/Project/A2 (1.171s)
QUILL_WORKERS=4 ./nqueens 12
OK
NQueens(12) Time = 1.085619sec
```

Time Execution for Private Deque Implementation -



Additional Information -

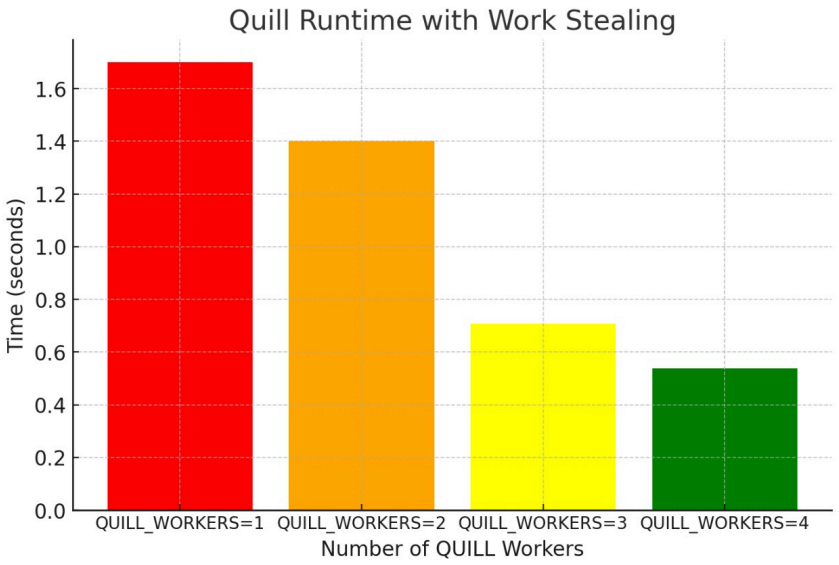
```
~/Desktop/SEM6/PRMP/Project/A2 (1.078s)
QUILL_WORKERS=4 ./nqueens 12

OK
NQueens(12) Time = 0.995749sec
Thread 0 executed 421242 tasks.
Thread 0 made 57 steal attempts.
Thread 0 successfully stole 0 tasks.
Thread 1 executed 382626 tasks.
Thread 1 made 24763 steal attempts.
Thread 1 successfully stole 9363 tasks.
Thread 2 executed 47808 tasks.
Thread 2 made 11 steal attempts.
Thread 2 successfully stole 1 tasks.
Thread 3 executed 4512 tasks.
Thread 3 made 15 steal attempts.
Thread 3 successfully stole 11 tasks.
```

Quill Runtime with Work Stealing (Deadline -1 Implementation) -

Execution Time (No Private Deque)-

```
~/Desktop/SEM6/PRMP/Project/A1 (1.769s)
QUILL_WORKERS=1 ./nqueens 12

OK
NQueens(12) Time = 1.700400sec

~/Desktop/SEM6/PRMP/Project/A1 (1.471s)
QUILL_WORKERS=2 ./nqueens 12

OK
NQueens(12) Time = 1.401061sec

~/Desktop/SEM6/PRMP/Project/A1 (0.779s)
QUILL_WORKERS=3 ./nqueens 12

OK
NQueens(12) Time = 0.708370sec

~/Desktop/SEM6/PRMP/Project/A1 (0.607s)
QUILL_WORKERS=4 ./nqueens 12

OK
NQueens(12) Time = 0.537758sec
```

Time Execution for Non-Private Deque Work Stealing Implementation -


Quill Runtime with Work Stealing
Time (seconds)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
QUILL_WORKERS=1
QUILL_WORKERS=2
QUILL_WORKERS=3
QUILL_WORKERS=4
Number of QUILL Workers